PAPER_1577_DIAMOND_MOHS_10

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PAPER_1577 — Diamond Mohs Hardness = Mohs = SO_5 = 10 (single primitive — cleanest possible closure)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Engineering
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Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S571 (Engineering Unified Proof Set) lists **Diamond Mohs Hardness** as an EXACT-tier UQFF closure:

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`Mohs = SO_5 = 10 (single primitive — cleanest possible closure)`
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UQFF Closed Identity

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`Mohs = SO_5 = 10 (single primitive — cleanest possible closure) EXACT`
``

Pure integer-primitive expression with zero fitted constants.

Cleanest Possible UQFF Closure Diamond Mohs hardness = SO_5 — the **single locked integer primitive itself**. No arithmetic, no combination, no correction terms. The decimal-scale integer that anchors the SO(5) symmetry group order directly equals the empirical hardness scale maximum.

Mohs scale is bounded 1-10. UQFF's prediction: SO_5 = 10. Diamond is the hardest known mineral. The match is structural and exact.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209Y_S571
- Related: PAPER_1494-1575 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "diamond_mohs_10"})`
